

Convex combinations

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A convex set is a set containing all convex combinations of its elements, *i.e.*, a set $K \subseteq \mathbb{R}^n$ satisfying $\text{conv}(K) = K$. In this note we present some basic properties of convex combinations. We first show that the convex hull $\text{conv}(S)$ of any set $S \subseteq \mathbb{R}^n$ is a convex set.

Theorem 1 (*The convex hull of a set is a convex set*)

For every set $S \subseteq \mathbb{R}^n$, its convex hull $\text{conv}(S)$ is a convex set. ■

Proof: We must show $\text{conv}(S)$ contains all convex combinations of its elements. Therefore let $\bar{x} = \lambda_1 x^1 + \dots + \lambda_k x^k$ be a convex combination of k points $x^1, \dots, x^k \in \text{conv}(S)$, where $\lambda_i \geq 0$ and $\sum_{i=1}^k \lambda_i = 1$. We must show $\bar{x} \in \text{conv}(S)$.

Let $i \in \{1, \dots, k\}$ be arbitrary. Since $x^i \in \text{conv}(S)$, the point x^i is a convex combination of points in S , *i.e.*, one may write $x^i = \sum_{j \in J_i} \delta_{i,j} e^{i,j}$, where J_i is a finite index set, $\delta_{i,j} \geq 0$, $\sum_{j \in J_i} \delta_{i,j} = 1$ and $e^{i,j} \in S$ for all $j \in J_i$. One now has:

$$\bar{x} = \lambda_1 x^1 + \dots + \lambda_k x^k = \sum_{i=1}^k \sum_{j \in J_i} (\lambda_i \delta_{i,j}) \cdot e^{i,j}$$

where $\lambda_i \delta_{i,j} \geq 0$ and $\sum_{i=1}^k \sum_{j \in J_i} (\lambda_i \delta_{i,j}) = \sum_{i=1}^k \lambda_i \sum_{j \in J_i} \delta_{i,j} = 1$. Hence $\bar{x}$ is a convex combination of the points $e^{i,j} \in S$ for $i \in \{1, \dots, k\}$ and $j \in J_i$, and therefore $\bar{x} \in \text{conv}(S)$. ■

A convex set is a set $K \subseteq \mathbb{R}^n$ satisfying $\text{conv}(K) = K$. Therefore, since Lemma 1 shows $\text{conv}(S)$ is a convex set for every $S \subseteq \mathbb{R}^n$, one has:

Corollary 1 (*The convex hull of a convex hull*)

For any set $S \subseteq \mathbb{R}^n$ one has $\text{conv}(\text{conv}(S)) = \text{conv}(S)$. ■

In the remainder we give an alternative definition of an extreme point: A point is extreme for a convex set exactly when it is *not* a convex combination of two other points in the set:

Theorem 2 (*Equivalent definition of an extreme point*)

Let $K \subseteq \mathbb{R}^n$ be a convex set. A point $x \in K$ is extreme for $K \iff x$ is not a convex combination of two points in $K \setminus \{x\}$. ■

We now prove Theorem 2. One direction is immediate: If $x \in K$ is extreme for K , then x is by definition not a convex combination of points in $K \setminus \{x\}$, and therefore, in particular, not a convex combination of two points in $K \setminus \{x\}$.

Therefore suppose $x \in K$ is not extreme for K . Hence x is a convex combination of, say, k points, so we may write $x = \lambda_1 x^1 + \dots + \lambda_k x^k$, where $\lambda_i > 0$, $\sum_{i=1}^k \lambda_i = 1$ and $x^1, \dots, x^k \in K \setminus \{x\}$. We must show that x is also a convex combination of two points in $K \setminus \{x\}$.

Let $y^1 := x^1$ and $y^2 := \frac{1}{1-\lambda_1}(x - \lambda_1 x^1)$. Clearly $x = \lambda_1 y^1 + (1 - \lambda_1)y^2$. Furthermore, since $x^1 \neq x$, one has $y^1 \neq x$ and $y^2 \neq x$.

We now argue $y^2 \in K$. From this it then follows that x is a convex combination of the two points $y^1, y^2 \in K \setminus \{x\}$, which completes the proof of Theorem 2. Indeed, one may write:

$$y^2 = \frac{1}{1 - \lambda_1}(x - \lambda_1 x^1) = \sum_{i=2}^k \frac{\lambda_i}{1 - \lambda_1} \cdot x^i,$$

where $\sum_{i=2}^k \frac{\lambda_i}{1 - \lambda_1} = \frac{1}{1 - \lambda_1} \cdot \sum_{i=2}^k \lambda_i = \frac{1 - \lambda_1}{1 - \lambda_1} = 1$. Hence y^2 is a convex combination of $x^2, \dots, x^k \in K$, and therefore $y^2 \in K$.